

The Islamia University Of Bahawalpur



(SDD)

Desktop-Based Attendance System using Face Recognition

Submitted By:

Nida

S25BINFT7M04037

Submitted To:

Rana Nazir Sab

Semester:

7(M_1) ADP

Department :

Information Technology

june ,2026

Table of Contents

- ❖ Introduction
- ❖ System Architecture
- ❖ Component Design
- ❖ Database Design
- ❖ User Interface Design
- ❖ Algorithm Design
- ❖ Module Descriptions
- ❖ References

1. Introduction

1.1 Purpose

This Software Design Document (SDD) provides the complete technical design specification for the Desktop-Based Attendance System using Face Recognition. It describes the system architecture, component design, database schema, algorithms, and user interface details for developers and evaluators.

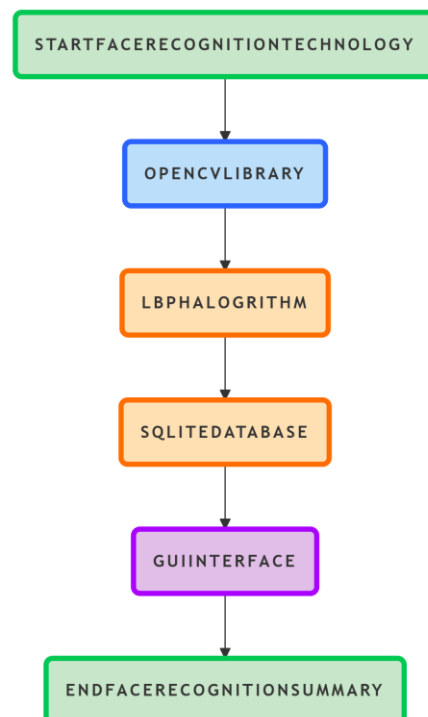
1.2 Scope

The document covers the architectural design, data structures, processing logic, and interface specifications for a desktop-based attendance system that operates offline using face recognition technology.

1.3 Definitions

Term	Definition
Face Recognition	Technology to identify a person using facial features
OpenCV	Open-source computer vision library
LBPH	Local Binary Patterns Histograms algorithm for face recognition
SQLite	Lightweight local database
GUI	Graphical User Interface

Diagram



2. System Architecture

2.1 Architecture Overview

The system follows a **two-tier architecture**:

Tier 1 (Client/Application Layer):

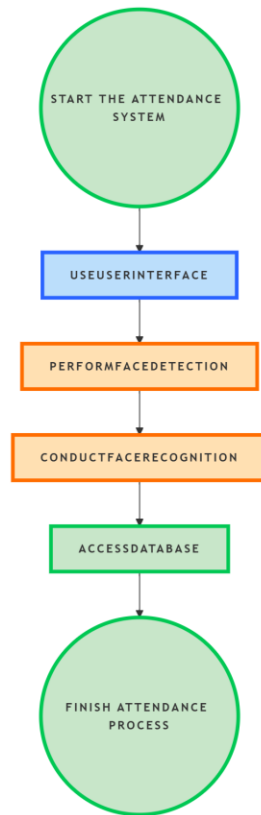
- ❖ Python-based desktop application
- ❖ GUI built with Tkinter/PyQt
- ❖ Face detection and recognition processing
- ❖

Tier 2 (Data Layer):

- ❖ SQLite local database
- ❖ Stores student information and attendance records
- ❖

2.2 System Components

Component	Technology	Purpose
User Interface	Tkinter/PyQt	Admin and teacher interaction
Face Detection	OpenCV (Haar Cascade)	Detect faces from webcam
Face Recognition	OpenCV (LBPH)	Recognize registered faces
Database	SQLite	Store student and attendance data



2.4 Directory Structure

text

AttendanceSystem/

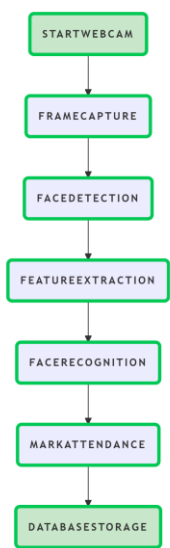
- |— main.py
- |— train.py
- |— attendance.py
- |— database.py
- |— dataset/
 - | └─ [StudentID]_[Name]_[image].jpg
- |— trainer/
 - | └─ trainer.yml
- |— attendance_records.csv

3. Component Design

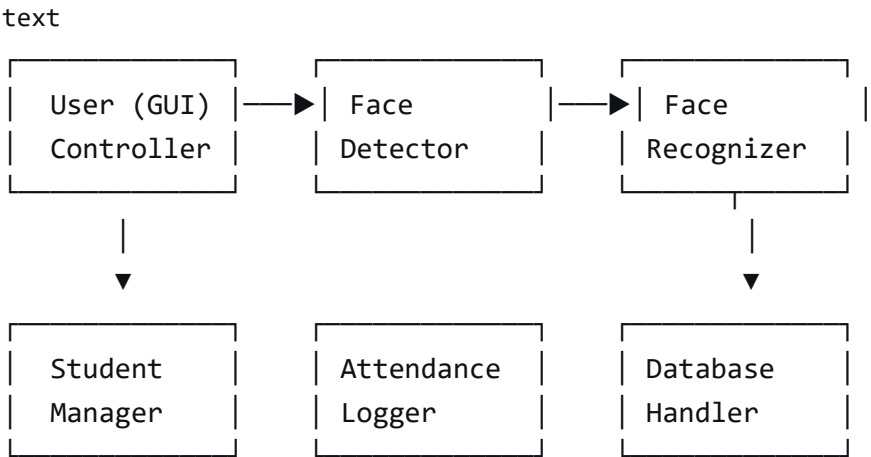
3.1 Component List

Component ID	Name	Responsibility
C1	Face Detector	Detect faces from webcam feed
C2	Face Trainer	Train LBPH model with dataset
C3	Face Recognizer	Recognize detected faces
C4	Attendance Logger	Record attendance with timestamp
C5	Student Manager	Add/delete student records
C6	Report Viewer	Display attendance records

2.3 Data Flow



3.2 Component Interaction



4. Database Design

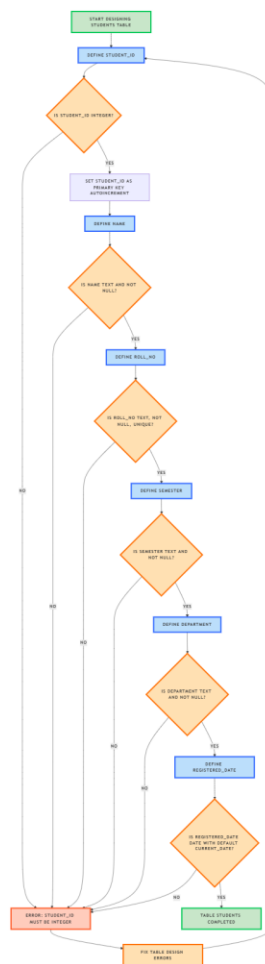
4.1 Database Schema

Database Name: attendance.db

4.2 Table: Students

Field	Type	Constraints	Description
student_id	INTEGER	PRIMARY KEY AUTOINCREMENT	Unique student identifier
name	TEXT	NOT NULL	Student full name
roll_no	TEXT	NOT NULL UNIQUE	University roll number

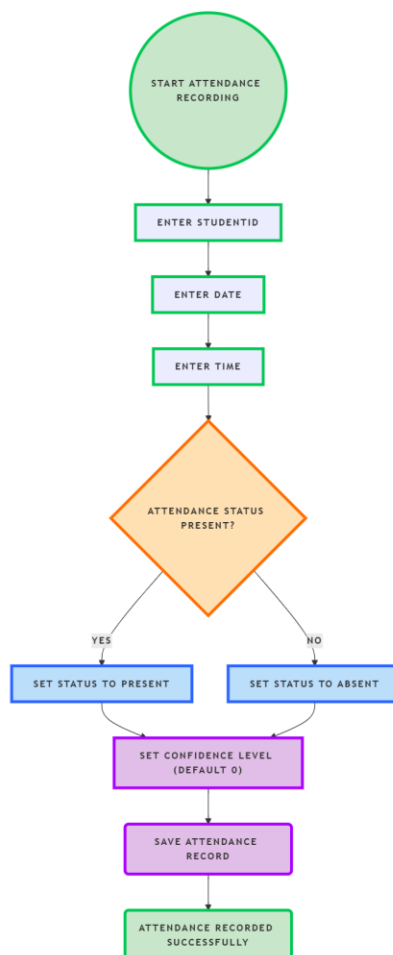
Field	Type	Constraints	Description
semester	TEXT	NOT NULL	Current semester
department	TEXT	NOT NULL	Department name
registered_date	DATE	DEFAULT CURRENT_DATE	Date of registration



4.3 Table: Attendance

Field	Type	Constraints	Description
attendance_id	INTEGER	PRIMARY KEY	Unique record

Field	Type	Constraints	Description
		AUTOINCREMENT	identifier
student_id	INTEGER	FOREIGN KEY (Students)	Reference to student
date	DATE	NOT NULL	Attendance date
time	TIME	NOT NULL	Attendance time
status	TEXT	NOT NULL	Present/Absent
recognized_confidence	INTEGER	DEFAULT 0	Recognition confidence level

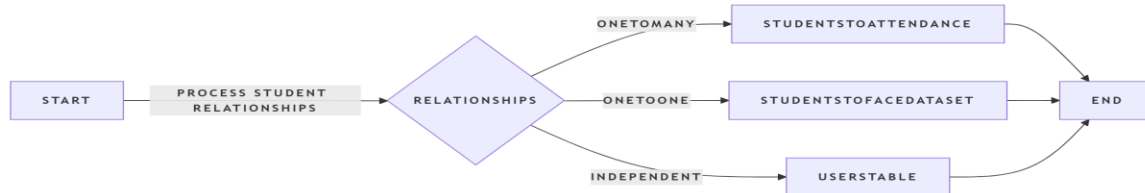


4.4 Table: Users

Field	Type	Constraints	Description
user_id	INTEGER	PRIMARY KEY AUTOINCREMENT	Unique user identifier
username	TEXT	NOT NULL UNIQUE	Login username
password	TEXT	NOT NULL	Encrypted password

4.5 Entity Relationship Diagram Description

- **Students** has **One-to-Many** relationship with **Attendance** (one student can have many attendance records)
- **Students** is referenced by **FaceDataset** (one-to-one relationship)
- **Users** table is independent with no direct relationship to other tables



5. User Interface Design

5.1 Interface Overview

Window Name	Components	Purpose
Main Window	Start Attendance, Add Student, View Records	Primary navigation
Add Student	Name fields, Roll No, Capture Button	Register new student
Attendance Window	Webcam feed, Start/Stop Button	Mark attendance
Reports Window	Table view, Export Button	View attendance records

5.2 Page Specifications

Main Window:

- Title: "Attendance System - Face Recognition"
- Buttons: Start Attendance, Add Student, View Records, Exit
- Status bar showing current time and webcam status

Add Student Window:

- Input fields: Name, Roll No, Semester, Department
- Action: Capture Images (20-30 images for training)

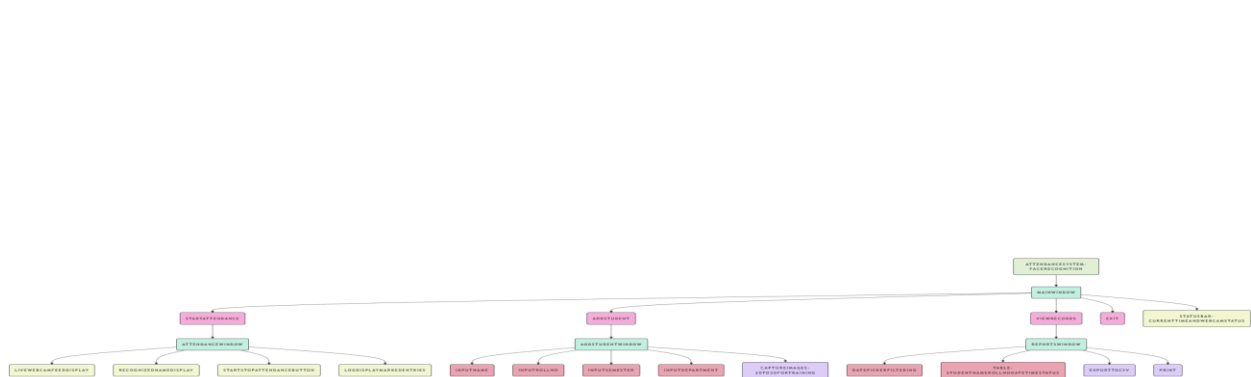
Attendance Window:

- Live webcam feed display
- Recognized name display
- Start/Stop attendance button
- Log display showing marked entries

Reports Window:

- Date picker for filtering
- Table showing: Student Name, Roll No, Date, Time, Status
- Export to CSV button
- Print button

Diagram



6. Algorithm Design

6.1 Face Detection Algorithm

Method: OpenCV Haar Cascade Classifier

Steps:

1. Load pre-trained Haar cascade XML file
2. Convert webcam frame to grayscale

3. Apply detectMultiScale() function
4. Return rectangle coordinates of detected faces

Parameters:

- scaleFactor: 1.1
- minNeighbors: 5
- minSize: 50x50 pixels

6.2 Face Recognition Algorithm

Method: LBPH (Local Binary Patterns Histograms)

Training Phase Steps:

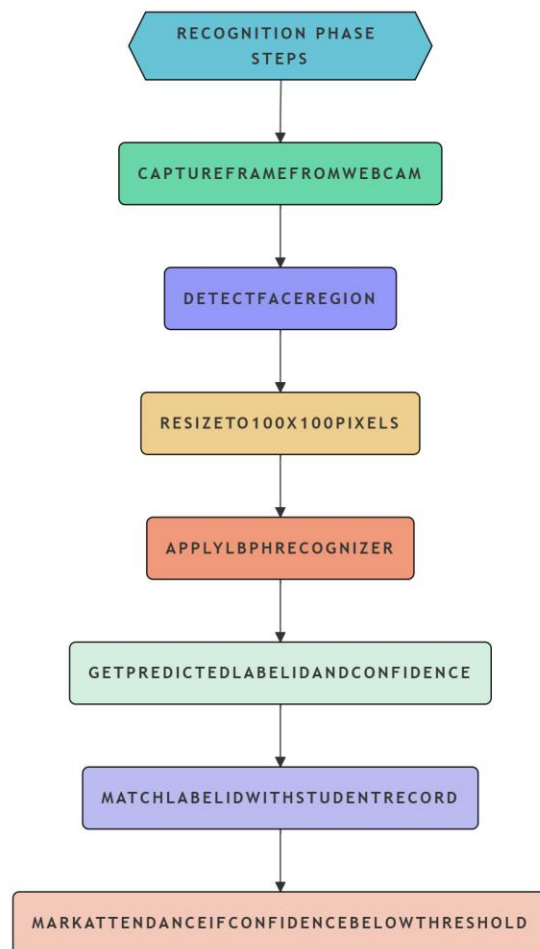
Step	Operation
1	Load all images from dataset folder
2	Convert each image to grayscale
3	Detect face region
4	Extract LBP features
5	Create label IDs for each person
6	Train recognizer with features and labels
7	Save trained model to trainer.yml

Recognition Phase Steps:

Step	Operation
1	Capture frame from webcam

Step	Operation
2	Detect face region
3	Resize to 100x100 pixels
4	Apply LBPH recognizer
5	Get predicted label ID and confidence
6	Match label ID with student record
7	Mark attendance if confidence < threshold

Confidence Threshold: 60 (below 60 = recognized, above = unknown)



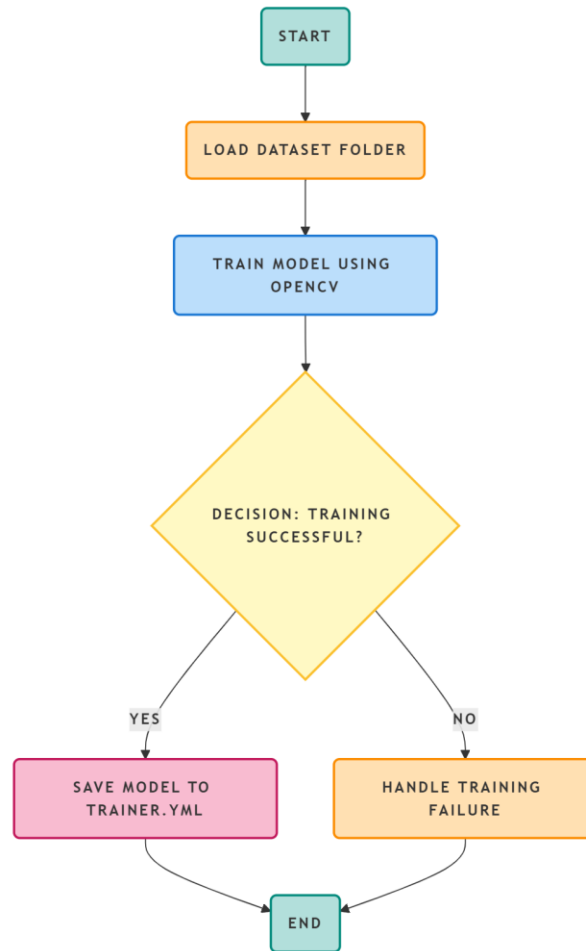
7. Module Descriptions

7.1 Module: Face Detection Module

Property	Value
Module Name	face_detector.py
Input	Webcam frame
Output	Face coordinates and ROI
Dependencies	OpenCV (cv2)
Function	detect_face(frame) → face_roi, coordinates

7.2 Module: Face Training Module

Property	Value
Module Name	trainer.py
Input	Dataset folder with student images
Output	trainer.yml model file
Dependencies	OpenCV, os
Function	train_model(dataset_path) → trainer.yml

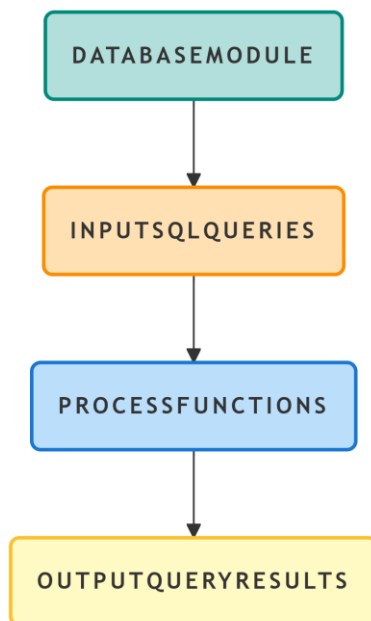


7.3 Module: Attendance Module

Property	Value
Module Name	attendance.py
Input	Recognized student ID
Output	Database record
Dependencies	SQLite3, datetime
Function	mark_attendance(student_id) → Boolean

7.4 Module: Database Module

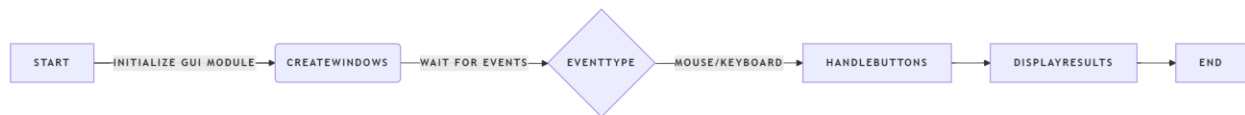
Property	Value
Module Name	database.py
Input	SQL queries
Output	Query results
Functions	add_student(), get_students(), save_attendance(), get_attendance()



7.5 Module: GUI Module

Property	Value
Module Name	main_gui.py
Input	User mouse/keyboard events

Property	Value
Output	Screen display
Dependencies	Tkinter/PyQt, PIL
Functions	create_windows(), handle_buttons(), display_results()



8. References

1. Bradski, G., "The OpenCV Library," Dr. Dobb's Journal of Software Tools, 2000.

2. Ahonen, T., Hadid, A., and Pietikainen, M., "Face Description with Local Binary Patterns," IEEE Transactions on Pattern Analysis and Machine Intelligence, 2006.
 3. OpenCV Documentation, "Face Recognition with OpenCV," [Online]. Available: .
 4. SQLite Documentation, "SQLite Database Library," [Online]. Available:
 5. Python Software Foundation, "Python Documentation," [Online]. Available:
-